

Project 1 — E-Commerce Sales Intelligence Dashboard

DAY 1

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Objective

Set up the complete development environment for Project 1 and verify that all 9 source CSV files from the Olist Brazilian E-Commerce dataset load correctly, producing zero missing-file errors. This is the foundation for the ETL pipeline to be run on Day 2.

Environment summary

Component	Details	Status
Operating system	Windows 11	OK
IDE	Visual Studio Code (latest)	OK
Python runtime	Anaconda (base) — Python 3.13	OK
Virtual env	Skipped — Anaconda base used directly	OK
Key packages	pandas, numpy (bundled with Anaconda)	OK
Extra packages	duckdb 1.5.3, openpyxl (pip install)	OK
Project location	C:\Users\Sudiksha Gunjkar\Study\VSCode\ecommerce-dashboard	OK

Steps completed

Install Python / verify Anaconda Done
Confirmed Anaconda Python 3.13 is available — (base) prefix visible in terminal.

2 **Create project folder structure** Done
Ran PowerShell commands to create ecommerce-dashboard\ with data\raw, data\processed, notebooks, sql subfolders.

Open folder in VS Code Done
Used File > Open Folder to set ecommerce-dashboard as the active workspace.
ECOMMERCE-DASHBOARD visible in Explorer.

4 **Set up Python environment** Done
Skipped venv (Anaconda conflict). Used Anaconda base directly. Ran pip install duckdb openpyxl.

Download Olist dataset from Kaggle Done
Downloaded archive.zip from kaggle.com/datasets/olistbr/brazilian-ecommerce. Extracted all 9 CSVs into data/raw/.

6 **Run sanity_check.py** Done
Ran python sanity_check.py — all 9 files loaded, row counts verified, nulls identified and documented.

Dataset verification — sanity_check.py output

All 9 CSV files confirmed present in [data/raw/](#). Row counts and null summary below:

File	Rows	Key columns	Nulls found
olist_orders_dataset	99,441	order_id, customer_id, status, timestamps	order_delivered_customer_date (2,965)
olist_order_items_dataset	112,650	order_id, product_id, seller_id, price, freight	None
olist_customers_dataset	99,441	customer_id, zip_code, city, state	None
olist_products_dataset	32,951	product_id, category_name, weight, dimensions	category_name (610), dimensions (2)
olist_sellers_dataset	3,095	seller_id, zip_code, city, state	None
olist_order_payments_dataset	103,886	order_id, payment_type, installments, value	None
olist_order_reviews_dataset	99,224	order_id, review_score, comment_title, message	comment_title (87,656), message (58,247)
olist_geolocation_dataset	1,000,163	zip_prefix, lat, lng, city, state	None
product_category_name_translation	71	category_name, category_name_english	None

Sanity check result: All 9 files found. You are ready for Day 2.

Total source records across all files: ~1.6 million rows

Challenges faced and fixes applied

#	Challenge	Root cause	Fix applied
1	&& operator not recognised in PowerShell	VS Code terminal defaulted to PowerShell. The && separator is a bash/cmd syntax — PowerShell uses ; or separate lines.	Ran each command on a separate line. mkdir accepted comma-separated targets: mkdir data\raw, data\processed, notebooks, sql
2	Folders created at wrong level (VSCode root instead of inside ecommerce-dashboard)	The cd into ecommerce-dashboard was part of the failed && chain, so it never executed. mkdir ran in the parent VSCode folder.	Re-ran cd into ecommerce-dashboard first, then ran mkdir. Verified with dir — all 4 folders visible inside the correct parent.
3	python -m venv venv failed with Anaconda traceback	Anaconda ships its own venv module which conflicts with standard venv creation when run from the Anaconda base environment.	Skipped venv entirely. Used Anaconda base environment directly. All required packages already present; only duckdb and openpyxl needed pip install.
4	review_comment_title and review_comment_message have large null counts	Reviews dataset: 87,656 null titles and 58,247 null messages out of 99,224 rows. Most buyers do not leave written comments on Olist.	This is expected and documented. ETL pipeline already handles it with fillna("") — no action needed beyond confirmation.

Key learnings from Day 1

1. PowerShell requires commands on separate lines — && chaining is a bash/cmd pattern that does not work in PS.
2. Anaconda users should skip `python -m venv` entirely and work in the Anaconda base environment or use `conda create` instead.
3. Large null counts in review text columns are a business reality (most users don't write comments), not a data quality problem.
4. DuckDB 1.5.3 installed cleanly alongside Anaconda via pip — no conflicts with existing packages.
5. Always verify the working directory (`pwd`) before running `mkdir` to avoid creating folders in the wrong location.

Day 2 — next steps

Task	Detail
Place <code>etl_pipeline.py</code> in the project root	<code>C:\Users\Sudiksha</code> <code>Gunjkar\Study\VSCode\ecommerce-dashboard\etl_pipeline.py</code>
Run the ETL pipeline	<code>python etl_pipeline.py</code> – expected runtime ~30-60 seconds
Read <code>etl_log.txt</code>	Review every <code>WARNING</code> line – document what caused it
Verify 6 output CSVs in <code>data/processed/</code>	<code>dim_date</code> , <code>dim_customer</code> , <code>dim_product</code> , <code>dim_seller</code> , <code>dim_geography</code> , <code>fact_orders</code>
Confirm row counts	<code>fact_orders</code> ~96k rows <code>dim_customer</code> ~99k <code>dim_product</code> ~32k